

Physics 514, Problem Set 5: General Relativity

due: *Tuesday, March 1, Noon*

Please note that, as you have two weeks to complete this problem set, it is a bit longer than normal; I suggest that you do not leave it until the night before! Please place your solutions in the box outside my office before the due date. You are encouraged to discuss these problems with your colleagues, but you must write up your own solutions; the solutions you hand in should reflect your own work and understanding.

Reading: Chapters 1, 2, 3 & 4 of Carroll.

1. Using Riemann normal coordinates, prove the Bianchi identity $\nabla_{[\lambda} R_{\rho\sigma]\mu\nu} = 0$. Using the Bianchi identity show that the Einstein tensor obeys $\nabla^\mu G_{\mu\nu} = 0$.
2. In general relativity we often consider an interesting sort of matter called "vacuum energy" whose stress tensor is proportional to the metric. The equations of motion are

$$G_{\mu\nu} = -\Lambda g_{\mu\nu}$$

where Λ is a parameter known as the cosmological constant. We will explore solutions to this equation of motion.

- (a) Show that the Ricci scalar of any solution of this equation is a constant, so that the equation of motion is equivalent to

$$R_{\mu\nu} = C g_{\mu\nu}$$

where C is a constant you should evaluate. A space-time solving this equation for some constant C is called an "Einstein space". The symmetric spaces we discussed last problem set are examples of Einstein spaces. We will now see that the four dimensional versions of these spaces are solutions of Einstein's equations with a cosmological constant.

- (b) Consider the four dimensional spacetime with coordinates (x^0, x^1, x^2, r) and metric

$$ds^2 = \ell^2 \frac{dr^2 - dx^{02} + dx^{12} + dx^{22}}{r^2}$$

This spacetime is known as Anti-de Sitter space (AdS). Show that if $\Lambda < 0$ then AdS is a solution to our equations of motion for a particular value of ℓ . AdS is the prototypical symmetric space-time with negative curvature; it can be thought of as the Lorentzian version of the hyperboloid.

- (c) Consider the four dimensional space-time with coordinates (t, x^1, x^2, x^3) and metric

$$ds^2 = -dt^2 + e^{2Ht}(dx^{1^2} + dx^{2^2} + dx^{3^2})$$

This spacetime is known as de Sitter space (dS). Show that if $\Lambda > 0$ then dS is a solution to our equations of motion for a particular value of H . dS is the prototypical symmetric space-time with positive curvature; it can be thought of as the Lorentzian version of the sphere.

- (d) The metric for dS describes a universe where the spatial directions are expanding at an exponential rate. The constant H is known as the *Hubble constant* of de Sitter space and describes the rate of this expansion. What are the dimensions of H ? For the above metric, how long (measured in terms of the time coordinate t) does it take space to double in size?
- (e) Show that Λ/G has units of energy density, i.e. Energy per unit length cubed (recall that we use units where $c = 1$). A typical particle physics energy scale is something like 1 GeV. Since Planck's constant has units of Energy times length this means that a typical particle physics energy density is $(1 \text{ GeV})^4/\hbar^3$. In quantum computations of the vacuum energy density we typically get answers of this size (though the precise answer depends on the details of the particle physics model). If we take the cosmological constant Λ/G to be this size, what is the corresponding Hubble constant? How long would it take for the universe to double in size? Astronomical observations indicate that the universe is indeed expanding, and doubles in size every few billion years. What is the corresponding Hubble constant? How does this compare to your previous result? Congratulations, you have just discovered the cosmological constant problem!
3. Consider a spacetime with metric $g_{\mu\nu}$ in coordinates x^μ . Consider a second metric $\hat{g}_{\mu\nu}$ which is related to the first metric by multiplication by some function $\omega(x^\mu)$

$$\hat{g}_{\mu\nu} = \omega(x^\mu)g_{\mu\nu}$$

We say that the metric $\hat{g}_{\mu\nu}$ is related to $g_{\mu\nu}$ by a *conformal transformation*, or that the two geometries are *conformally related*. We require $\omega(x^\mu) > 0$ whenever $g_{\mu\nu}$ is non-singular, since we need the metric to be invertible.

- (a) If we have two vectors v^μ and w^μ we can define the angle θ between these vectors by generalizing the formula from vector calculus

$$\cos \theta = \frac{g_{\mu\nu} v^\mu w^\nu}{\sqrt{v^2 w^2}}$$

where $v^2 = g_{\mu\nu} v^\mu v^\nu$. Convince yourself that for Euclidean three dimensional space this is the definition of angle that you are used to. Show that conformal transformations preserve angles between vectors. Show that a worldline which is timelike (or null) with respect to the original metric $g_{\mu\nu}$ is also timelike (or null) with respect to $\hat{g}_{\mu\nu}$. This means that if we are only trying to study the *causal structure* of spacetime (i.e. we are only asking whether two events can be connected by a timelike worldline) then we do not care about overall factors in front of the metric!

- (b) A spacetime which is conformally related to Minkowski space is called *conformally flat*. Show that AdS and dS are both conformally flat. (*Hint: for dS find a new time coordinate $\tau(t)$ which makes this obvious*).
4. Einstein's equations completely fix $R_{\mu\nu}$, which is the trace of the Riemann tensor. It is therefore convenient to define the Weyl tensor $C_{\mu\nu\rho\sigma}$, which is the Riemann tensor with all of the "trace" information removed:

$$C_{\mu\nu\rho\sigma} = R_{\mu\nu\rho\sigma} - \frac{2}{n-2} (g_{\rho[\mu} R_{\nu]\sigma} - g_{\sigma[\mu} R_{\nu]\rho}) + \frac{2}{(n-1)(n-2)} g_{\rho[\mu} g_{\nu]\sigma} R$$

where n is the dimension of the manifold.

- (a) Show that the Weyl tensor has the same symmetries as the Riemann tensor under permutation of the indices.
- (b) Show that if we trace over any two indices of the Weyl tensor (by raising an index and contracting with another) we get zero.
- (c) Using these facts, argue that the Weyl tensor has 10 degrees of freedom in four dimensions.

- (d) Show that any spacetime which is conformally flat has vanishing Weyl tensor. It turns out that in dimension $n \geq 4$ the converse is true as well: a spacetime is conformally flat if and only if the Weyl tensor vanishes. In fact, one can show that the Weyl tensor is always unchanged under a conformal transformation, although the computation is somewhat lengthy.
5. In class we showed that in four dimensions the Riemann tensor has 20 degrees of freedom while the Ricci tensor has only 10 degrees of freedom. How many degrees of freedom are contained in the Riemann and Ricci tensors in three dimensions? You should find that they contain the same number of degrees of freedom, which means that in three dimensions Einstein's equations completely fix the curvature of spacetime. Show that the Weyl tensor has zero degrees of freedom in three dimensions; this means that the Weyl tensor vanishes. This allows us to write the Riemann tensor explicitly in terms of the Ricci tensor!

Problem 1

1. Using Riemann normal coordinates, prove the Bianchi identity $\nabla_{[\lambda} R_{\rho\sigma]\mu\nu} = 0$. Using the Bianchi identity show that the Einstein tensor obeys $\nabla^\mu G_{\mu\nu} = 0$.

In general,

$$R^\rho_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\sigma\nu} + \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\nu\sigma} - (\mu \leftrightarrow \nu)$$

In RNC the first term is:

$$\begin{aligned} \partial_\mu \Gamma^\rho_{\sigma\nu} &= \frac{1}{2} \partial_\mu g^{\rho\alpha} (\partial_\nu g_{\alpha\sigma} + \partial_\sigma g_{\nu\alpha} - \partial_\alpha g_{\sigma\nu}) \\ &= \frac{1}{2} g^{\rho\alpha} (\partial_\mu \partial_\nu g_{\alpha\sigma} + \partial_\mu \partial_\sigma g_{\nu\alpha} - \partial_\mu \partial_\alpha g_{\sigma\nu}) \end{aligned}$$

The second term simply vanishes since in RNC $\Gamma = 0$ at the origin.

It follows that:

$$\begin{aligned} R_{\rho\sigma\mu\nu} &= \frac{1}{2} (\partial_\mu \partial_\nu g_{\rho\sigma} + \partial_\mu \partial_\sigma g_{\nu\rho} - \partial_\mu \partial_\rho g_{\sigma\nu}) \\ &\quad - \frac{1}{2} (\partial_\nu \partial_\mu g_{\rho\sigma} + \partial_\nu \partial_\sigma g_{\mu\rho} - \partial_\nu \partial_\rho g_{\sigma\mu}) \\ &= \frac{1}{2} (\partial_\mu \partial_\sigma g_{\nu\rho} - \partial_\mu \partial_\rho g_{\sigma\nu} + \partial_\nu \partial_\sigma g_{\mu\rho} - \partial_\nu \partial_\rho g_{\sigma\mu}) \end{aligned}$$

Now,

$$\nabla_{\tau\lambda} R_{\rho\sigma\gamma\mu\nu} = \frac{1}{3} (\nabla_\lambda R_{\rho\sigma\mu\nu} + \nabla_\rho R_{\sigma\lambda\mu\nu} + \nabla_\sigma R_{\lambda\rho\mu\nu})$$

where I've used the anti-symmetry of R 's first two indices.

Dropping the prefactors for now, & noting in RNC that ∇_λ is equivalent to ∂_λ , we get:

$$\begin{aligned} & \partial_\lambda (\underbrace{\partial_\mu \partial_\sigma g_{\nu\rho}}_{\text{yellow}} - \underbrace{\partial_\mu \partial_\rho g_{\sigma\nu}}_{\text{red}} + \underbrace{\partial_\nu \partial_\sigma g_{\mu\rho}}_{\text{pink}} - \underbrace{\partial_\nu \partial_\rho g_{\sigma\mu}}_{\text{green}}) \\ & + \partial_\rho (\underbrace{\partial_\mu \partial_\lambda g_{\nu\sigma}}_{\text{red}} - \underbrace{\partial_\mu \partial_\sigma g_{\lambda\nu}}_{\text{purple}} + \underbrace{\partial_\nu \partial_\lambda g_{\mu\sigma}}_{\text{green}} - \underbrace{\partial_\nu \partial_\sigma g_{\lambda\mu}}_{\text{blue}}) \\ & + \partial_\sigma (\underbrace{\partial_\mu \partial_\rho g_{\nu\lambda}}_{\text{purple}} - \underbrace{\partial_\mu \partial_\lambda g_{\rho\nu}}_{\text{yellow}} + \underbrace{\partial_\nu \partial_\rho g_{\mu\lambda}}_{\text{blue}} - \underbrace{\partial_\nu \partial_\lambda g_{\rho\mu}}_{\text{pink}}) \end{aligned}$$

$$= 0 \quad \text{where color-coded terms cancel.}$$

Thus: $\nabla_{\tau\lambda} R_{\rho\sigma\gamma\mu\nu} = 0$ as required.

Consider the form:

$$\nabla_\lambda R_{\rho\sigma\mu\nu} + \nabla_\rho R_{\sigma\lambda\mu\nu} + \nabla_\sigma R_{\lambda\rho\mu\nu} = 0.$$

$$\Rightarrow -\nabla_\lambda R_{\mu\nu\rho\sigma} + \nabla_\rho R_{\mu\nu\sigma\lambda} + \nabla_\sigma R_{\nu\mu\rho\lambda} = 0.$$

Now, contract μ & σ , ν & ρ to get:

$$-\nabla_\lambda R^{\sigma\rho}{}_{\sigma\rho} + \nabla^\rho R^\sigma{}_{\rho\sigma\lambda} + \nabla^\sigma R^\rho{}_{\sigma\rho\lambda} = 0.$$

$$\Rightarrow 2\nabla^\sigma R_{\sigma\lambda} - \nabla_\lambda R = 0.$$

$$\Rightarrow \nabla^\mu (R_{\mu\nu} - \frac{1}{2}R g_{\mu\nu}) = 0.$$

$$\Rightarrow \nabla^\mu G_{\mu\nu} = 0 \quad \text{as required.}$$

Problem 2

We solve:

$$G_{\mu\nu} = -\Lambda g_{\mu\nu}$$

a. Contracting both sides, assuming n -dimensional space,

$$R - \frac{1}{2}nR = -\Lambda n$$

$$\Rightarrow R = \frac{2n}{n-2} \Lambda$$

In particular, $n=4$ yields $R=4\Lambda$.

Plugging this back in,

$$R_{\mu\nu} = \frac{2}{n-2} \Lambda g_{\mu\nu}$$

$$= \frac{1}{n} R g_{\mu\nu}$$

b. As usual, we compute $R^\rho{}_{\sigma\mu\nu}$.

I'm just going to skip to the results.

The metric:

$$ds^2 = \frac{l^2}{r^2} (-dt^2 + dx^2 + dy^2 + dz^2)$$

We find that : (computations below)

$$R_{\mu\nu} = -\frac{3}{l^2} \eta_{\mu\nu}$$

This is a solution if :

$$-\frac{3}{l^2} = \frac{l^2}{l^2} \Lambda$$

$$\Rightarrow l = \left(-\frac{3}{\Lambda}\right)^{1/2}$$

(noting that $\Lambda < 0$)

c. The metric is now:

$$ds^2 = -dt^2 + e^{2Ht}(dx^2 + dy^2 + dz^2)$$

which gives:

(again, see below).

$$R_{\mu\nu} = 3H^2 g_{\mu\nu}$$

This is a solution if :

$$\Lambda = 3H^2 \Rightarrow H = \left(\frac{\Lambda}{3}\right)^{1/2}$$

(assume $\Lambda > 0$).

d. Ht is dimensionless, so it has units of $(\text{time})^{-1}$, which is also $(\text{length})^{-1}$ if $c=1$, & also (energy) if $\hbar=1$.

Space doubles in size every Δt , where we require

$$e^{2H\Delta t} = 2$$

$$\Rightarrow \Delta t = \frac{1}{2H} \ln 2$$

e. Λ has units $(\text{length})^{-2}$ since it's proportional to the Ricci scalar. G has units of $\text{m}^3 \text{kg}^{-1} \text{s}^{-2} \equiv \text{mkg}^{-1}$ since $c=1$. It is then clear that Λ/G has units kg/m^3 , but $[E] = [m]$ so this is an energy density.

We get:

$$\Lambda = \frac{(1 \text{ GeV})^4}{\hbar^3} G$$

$$\Rightarrow H = (1 \text{ GeV})^2 \left(\frac{G}{3\hbar^3} \right)^{1/2} c^{-2.5} = 71800 \text{ s}^{-1}$$

$$\text{In this case, } \Delta t = 4.8 \times 10^{-6} \text{ s}$$

$\sim 10^9$ years
 $\nearrow$

Assuming instead that $\Delta t \sim 10^{16}$ s, we see that:

$$H = 3.5 \times 10^{-17} \text{ s}^{-1}$$

Off by about 22 orders of magnitude.

Problem 3

a. Normally we define:

$$\cos \theta \equiv \frac{\vec{v} \cdot \vec{w}}{vw}$$

In $\mathbb{R}^3$ we can write this tensorially as:

$$\cos \theta \equiv \frac{\delta_{ij} v^i w^j}{\sqrt{v^2 w^2}}$$

with $v^2 = \delta_{ij} v^i v^j$. Replacing δ_{ij} with a curved space metric gives the desired result.

Under a conformal transformation,

$$\cos \hat{\theta} = \frac{\omega g_{\mu\nu} v^\mu w^\nu}{\sqrt{\omega v^2 \omega w^2}} = \cos \theta$$

Since $\omega(x^\mu)$ is strictly positive whenever g is non-zero, this factor of ω does not change the sign of $v^\mu v_\mu$. So it preserves the fact that v is timelike or null.

b. This is obvious for AdS.

ds has the metric:

$$ds^2 = -dt^2 + e^{2Ht} (dx^2 + dy^2 + dz^2)$$

To show ds is conformally flat, we suppose the metric under some transformation $t \rightarrow \tau$ takes the form:

$$ds^2 = e^{2H\tau} (-d\tau^2 + dx^2 + dy^2 + dz^2)$$

where of course e^{2Ht} can be rewritten in terms of τ post-facto.

This need not work, but it would if we could solve:

see Carroll p. 77

$$dt^2 = e^{2Ht} d\tau^2$$

$$\Rightarrow \frac{d\tau}{dt} = e^{-Ht}$$

$$\Rightarrow \tau = -\frac{1}{H} e^{-Ht}$$

$$\Rightarrow e^{2Ht} = \frac{H^2}{\tau^2}$$

So that:

$$ds^2 = \frac{H^2}{\tau^2} (-d\tau^2 + dx^2 + dy^2 + dz^2)$$

Now it is clear ds is conformally flat.

Problem 4

a.
$$C_{\mu\nu\rho\sigma} = R_{\mu\nu\rho\sigma} - \frac{2}{n-2} (g_{\rho[\mu} R_{\nu]\sigma} - g_{\sigma[\mu} R_{\nu]\rho}) + \frac{2}{(n-1)(n-2)} g_{\rho[\mu} g_{\nu]\sigma} R$$

The Riemann tensor has four symmetries to look for:

$$R_{\mu\nu\rho\sigma} = -R_{\nu\mu\rho\sigma} \quad (1)$$

$$R_{\mu\nu\rho\sigma} = -R_{\mu\nu\sigma\rho} \quad (2)$$

$$R_{\mu\nu\rho\sigma} = R_{\rho\sigma\mu\nu} \quad (3)$$

$$R_{[\mu\nu\rho\sigma]} = 0 \quad (4)$$

Sometimes the last one is replaced with:

$$R_{\mu[\nu\rho\sigma]} = 0,$$

which is a stronger statement. But together with other symmetries, picking one or the other makes no difference.

(1) is trivial. (2) is also easy if we expand out the antisymmetrization of the

last term:

$$g_{\rho\tau\mu} g_{\nu\sigma} = \frac{1}{2} (g_{\rho\mu} g_{\nu\sigma} - g_{\rho\nu} g_{\mu\sigma})$$

which is indeed antisymmetric under $\rho \leftrightarrow \sigma$.
Incidentally it's also obvious it satisfies (3).
So to prove (3), we need only worry about the middle term. Expanding it out:

$$g_{\rho\mu} R_{\nu\sigma} - g_{\rho\nu} R_{\mu\sigma} - g_{\sigma\mu} R_{\nu\rho} + g_{\sigma\nu} R_{\mu\rho}$$

Using the symmetry of $g_{\mu\nu}$ & $R_{\mu\nu}$, (3) is clear.

Finally, the LHS of (4) gives:

$$C_{[\mu\nu\rho\sigma]} = R_{[\mu\nu\rho\sigma]} - \frac{2}{n-2} (g_{[\rho\mu} R_{\nu\sigma]} - g_{[\sigma\mu} R_{\nu\rho]}) \\ + \frac{2}{(n-1)(n-2)} g_{[\rho\mu} g_{\nu\sigma]} R$$

$R_{[\mu\nu\rho\sigma]} = 0$, of course, & the other terms vanish too because they take antisymmetrizations over symmetric indices.

b. There are 6 possible contractions:

1. $C^\mu_{\mu\rho\sigma}$

2. $C^\mu_{\rho\mu\sigma}$

3. $C^\mu_{\rho\sigma\mu}$

4. $C_\rho{}^\mu{}_{\mu\sigma}$

5. $C_\rho{}^\mu{}_{\sigma\mu}$

6. $C_{\rho\sigma}{}^\mu{}_\mu$

Going in order:

1. $C^\mu_{\mu\rho\sigma} = g^{\mu\nu} C_{\nu\mu\rho\sigma} = -g^{\mu\nu} C_{\mu\nu\rho\sigma} = -C^\mu_{\mu\rho\sigma}$

$$\Rightarrow C^\mu_{\mu\rho\sigma} = 0.$$

2. $C^\mu_{\nu\mu\sigma} = g^{\mu\rho} C_{\mu\nu\rho\sigma}$

$$\begin{aligned} &= R_{\nu\sigma} - \frac{2}{n-2} \frac{1}{2} (n R_{\nu\sigma} - R_{\nu\sigma} - R_{\nu\sigma} + g_{\nu\sigma} R) \\ &\quad + \frac{2}{(n-1)(n-2)} \frac{1}{2} (n g_{\nu\sigma} - g_{\nu\sigma}) R \end{aligned}$$

$$= 0.$$

$$3. C^{\mu}{}_{\rho\sigma\mu} = -C^{\mu}{}_{\nu\mu\sigma} = 0.$$

$$4. C^{\mu}{}_{\rho}{}^{\mu}{}_{\sigma} = -C^{\mu}{}_{\rho\mu\sigma} = 0.$$

$$5. C^{\mu}{}_{\rho}{}^{\sigma}{}_{\mu} = -C^{\mu}{}_{\rho}{}^{\mu}{}_{\sigma} = 0.$$

$$6. C^{\mu}{}_{\rho\sigma}{}^{\mu} = C^{\mu}{}_{\mu\rho\sigma} = 0.$$

c. The Weyl tensor is constrained to 20 DOFs just by virtue of its shared symmetries with the Riemann tensor.

But it has an additional constraint:

$$C^{\mu}{}_{\rho\mu\sigma} = 0$$

This is obviously untrue for the Riemann tensor, since this is how the Ricci tensor is constructed.

Numbers 1 & 6 in part (c) were already true for the Riemann tensor, & the rest are the same as 2. So this is the only new constraint.

d. Go to the "Appendices" below.

Problem 5

The Riemann tensor, $R_{\mu\nu\rho\sigma}$, has symmetries (1) - (4).

That μ, ν are antisymmetric give 3 DOFs for a given ρ, σ . The same goes the other way: there are 3 DOFs in the ρ, σ indices for a given μ, ν . Normally we just multiply them, but here, (μ, ν) & (ρ, σ) are symmetric under exchange.

If we think of $\mu\nu, \rho\sigma$ each as one index that runs from 1-3, & then assume symmetry under exchange, the total DOFs is just the same as the DOFs of a symmetric 3×3 matrix.

Namely, 6 DOFs. In 3-D, (4) is satisfied automatically since at least one index must repeat. The Ricci tensor is a symmetric 3×3 matrix so it also has 6 DOFs.

The Weyl tensor has the additional condition:

$$C^{\rho}{}_{\mu\rho\nu} = 0.$$

Since μ, ν are symmetric, this represents 6 more constraints, leaving the Weyl tensor with zero degrees of freedom.

"Appendices" $\rightarrow$ work that looks like shit.

4 (d)

Weyl tensor conformal transformation. $\star$

$$\Gamma_{\mu\nu}^{\rho} = \frac{1}{2} g^{\rho\sigma} (\partial_{\nu} g_{\sigma\mu} + \partial_{\mu} g_{\nu\sigma} - \partial_{\sigma} g_{\mu\nu})$$
$$= \frac{1}{2} \omega \eta^{\rho\sigma} (\eta_{\sigma\mu} \partial_{\nu} \omega + \eta_{\nu\sigma} \partial_{\mu} \omega - \eta_{\mu\nu} \partial_{\sigma} \omega)$$

$$= \frac{1}{2} \omega (\delta_{\mu}^{\rho} \partial_{\nu} \omega + \delta_{\nu}^{\rho} \partial_{\mu} \omega - \eta^{\rho\sigma} \eta_{\mu\nu} \partial_{\sigma} \omega)$$

$\downarrow$ tensorial?

$$R^{\rho}_{\sigma\mu\nu} = \partial_{\mu} \Gamma_{\nu\sigma}^{\rho} - \partial_{\nu} \Gamma_{\mu\sigma}^{\rho} + \Gamma_{\mu\lambda}^{\rho} \Gamma_{\nu\sigma}^{\lambda} - \Gamma_{\nu\lambda}^{\rho} \Gamma_{\mu\sigma}^{\lambda}$$

$$R_{\rho\sigma\mu\nu} = g_{\rho\alpha} R^{\alpha}_{\sigma\mu\nu}$$

Riemann normal coord.: $\partial_\sigma g_{\mu\nu} = \eta_{\mu\nu} \partial_\sigma \omega \approx 0$

$$R^\alpha_{\sigma\mu\nu} \quad g_{\mu\nu} = \omega \eta_{\mu\nu} = \eta_{\mu\nu} \quad \xrightarrow{\omega=1} \quad \partial_\sigma \omega = 0.$$

$$= \partial_\mu \left(\frac{1}{2} \omega \left(\delta^\alpha_\nu \partial_\sigma \omega + \cancel{\delta^\alpha_\sigma \partial_\nu \omega} - \eta^{\alpha\rho} \eta_{\nu\sigma} \partial_\rho \omega \right) \right)$$

$$- \partial_\nu \left(\frac{1}{2} \omega \left(\delta^\alpha_\mu \partial_\sigma \omega + \cancel{\delta^\alpha_\sigma \partial_\mu \omega} - \eta^{\alpha\rho} \eta_{\mu\sigma} \partial_\rho \omega \right) \right)$$

$$= \frac{1}{2} \left(\delta^\alpha_\nu \partial_\mu \partial_\sigma \omega - \eta^{\alpha\beta} \eta_{\nu\sigma} \partial_\mu \partial_\beta \omega \right.$$

$$\left. - \delta^\alpha_\mu \partial_\nu \partial_\sigma \omega + \eta^{\alpha\beta} \eta_{\mu\sigma} \partial_\nu \partial_\beta \omega \right)$$

$$\Leftrightarrow R_{\rho\sigma\mu\nu}$$

$$= \frac{1}{2} \left(\eta_{\rho\nu} \partial_\mu \partial_\sigma \omega - \delta^\rho_\beta \eta_{\nu\sigma} \partial_\mu \partial_\beta \omega \right.$$

$$\left. - \eta_{\rho\mu} \partial_\nu \partial_\sigma \omega + \delta^\rho_\beta \eta_{\mu\sigma} \partial_\nu \partial_\beta \omega \right)$$

$$= \frac{1}{2} \left(\eta_{\rho\nu} \partial_\mu \partial_\sigma \omega - \eta_{\nu\sigma} \partial_\mu \partial_\rho \omega \right.$$

$$\left. - \eta_{\rho\mu} \partial_\nu \partial_\sigma \omega + \eta_{\mu\sigma} \partial_\nu \partial_\rho \omega \right)$$

$$C_{\mu\nu\rho\sigma} = R_{\mu\nu\rho\sigma} - \frac{2}{n-2} (g_{\rho[\mu} R_{\nu]\sigma} - g_{\sigma[\mu} R_{\nu]\rho}) + \frac{2}{(n-1)(n-2)} g_{\rho[\mu} g_{\nu]\sigma} R$$

$$R_{\sigma\nu} = R^{\rho}{}_{\sigma\rho\nu} = \eta^{\rho\mu} R_{\rho\sigma\mu\nu}$$

$$= \frac{1}{2} \left(\partial_\nu \partial_\sigma \omega - \eta_{\nu\sigma} \eta^{\rho\mu} \partial_\mu \partial_\rho \omega \right. \\ \left. - n \partial_\nu \partial_\sigma \omega + \partial_\nu \partial_\sigma \omega \right)$$

$$= \frac{1}{2} (2-n) \partial_\nu \partial_\sigma \omega - \frac{1}{2} \eta_{\nu\sigma} \eta^{\rho\mu} \partial_\mu \partial_\rho \omega$$

$$R = \eta^{\sigma\nu} R_{\sigma\nu} = \frac{1}{2} (2-n) \partial_\nu \partial_\sigma \omega - \frac{1}{2} \eta_{\nu\sigma} (\partial^2 \omega)$$

$$= \frac{1}{2} (2-n) \eta^{\nu\sigma} \partial_\nu \partial_\sigma \omega - \frac{1}{2} n \eta^{\nu\sigma} \partial_\nu \partial_\sigma \omega$$

$$= (1-n) \eta^{\nu\sigma} \partial_\nu \partial_\sigma \omega = (1-n) (\partial^2 \omega)$$

$$C_{\mu\nu\rho\sigma} = R_{\mu\nu\rho\sigma} - \frac{1}{n-2} \left(g_{\rho\mu} R_{\nu\sigma} - g_{\rho\nu} R_{\mu\sigma} \right. \\ \left. - g_{\sigma\mu} R_{\nu\rho} + g_{\sigma\nu} R_{\mu\rho} \right)$$

$$+ \frac{1}{(n-1)(n-2)} (g_{\rho\mu} g_{\nu\sigma} - g_{\rho\nu} g_{\mu\sigma}) R.$$

$$= \frac{1}{2} \left(\eta_{\rho\nu} \partial_\mu \partial_\sigma \omega - \eta_{\nu\sigma} \partial_\mu \partial_\rho \omega \right. \\ \left. - \eta_{\rho\mu} \partial_\nu \partial_\sigma \omega + \eta_{\mu\sigma} \partial_\nu \partial_\rho \omega \right) \\ - \frac{1}{n-2} \left(\eta_{\rho\mu} \left(\frac{1}{2} (2-n) \partial_\sigma \partial_\nu \omega - \frac{1}{2} \eta_{\sigma\nu} (\partial^2 \omega) \right) \right. \\ \left. - \eta_{\rho\nu} \left(\frac{1}{2} (2-n) \partial_\sigma \partial_\mu \omega - \frac{1}{2} \eta_{\sigma\mu} (\partial^2 \omega) \right) \right. \\ \left. - \eta_{\sigma\mu} \left(\frac{1}{2} (2-n) \partial_\rho \partial_\nu \omega - \frac{1}{2} \eta_{\rho\nu} (\partial^2 \omega) \right) \right. \\ \left. + \eta_{\sigma\nu} \left(\frac{1}{2} (2-n) \partial_\rho \partial_\mu \omega - \frac{1}{2} \eta_{\rho\mu} (\partial^2 \omega) \right) \right)$$

$$+ \frac{1}{(n-1)(n-2)} (\eta_{\rho\mu} \eta_{\nu\sigma} - \eta_{\rho\nu} \eta_{\mu\sigma}) (1-n) (\partial^2 \omega)$$

$$= -\frac{1}{n-2} \left(-\frac{1}{2} \eta_{\rho\mu} \eta_{\sigma\nu} + \frac{1}{2} \eta_{\rho\nu} \eta_{\sigma\mu} \right. \\ \left. + \frac{1}{2} \eta_{\sigma\mu} \eta_{\rho\nu} - \frac{1}{2} \eta_{\sigma\nu} \eta_{\rho\mu} \right) (\partial^2 \omega) \\ - \frac{1}{n-2} (\eta_{\rho\mu} \eta_{\nu\sigma} - \eta_{\rho\nu} \eta_{\mu\sigma}) (\partial^2 \omega)$$

$$\Rightarrow 0.$$

So C is zero in any code.

ADS space computation...

$$\Gamma_{\mu\nu}^t = \begin{pmatrix} 0 & 0 & 0 & -1/r \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1/r & 0 & 0 & 0 \end{pmatrix}$$

$$\Gamma_{\mu\nu}^x = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1/r \\ 0 & 0 & 0 & 0 \\ 0 & -1/r & 0 & 0 \end{pmatrix}$$

$$\Gamma_{\mu\nu}^y = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1/r \\ 0 & 0 & -1/r & 0 \end{pmatrix}$$

$$\Gamma_{\mu\nu}^{\lambda} = \begin{pmatrix} -1/r & 0 & 0 & 0 \\ 0 & 1/r & 0 & 0 \\ 0 & 0 & 1/r & 0 \\ 0 & 0 & 0 & -1/r \end{pmatrix}$$

$$R_{\mu\nu} = R^0_{\mu 0 \nu} + R^1_{\mu 1 \nu} + R^2_{\mu 2 \nu} + R^3_{\mu 3 \nu}$$

$$R^{\alpha}_{\sigma\mu\nu} = \partial_{\mu} \Gamma^{\alpha}_{\nu\sigma} - \partial_{\nu} \Gamma^{\alpha}_{\mu\sigma} + \Gamma^{\alpha}_{\mu\lambda} \Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\alpha}_{\nu\lambda} \Gamma^{\lambda}_{\mu\sigma}$$

$$\partial_0 \Gamma^0_{\nu\mu} - \partial_{\nu} \Gamma^0_{0\mu}$$

$$+ \Gamma^0_{0\lambda} \Gamma^{\lambda}_{\nu\mu} - \Gamma^0_{\nu\lambda} \Gamma^{\lambda}_{0\mu}$$

$$R^0_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & -1/\kappa^2 & 0 & 0 \\ 0 & 0 & -1/\kappa^2 & 0 \\ 0 & 0 & 0 & -1/\kappa^2 \end{pmatrix}$$

$$R^1_{\mu\nu} = \begin{pmatrix} 1/\kappa^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1/\kappa^2 & 0 \\ 0 & 0 & 0 & -1/\kappa^2 \end{pmatrix}$$

$$R^2_{\mu\nu} = \begin{pmatrix} 1/\kappa^2 & 0 & 0 & 0 \\ 0 & -1/\kappa^2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1/\kappa^2 \end{pmatrix}$$

$$R^3_{\mu\nu} = \begin{pmatrix} 1/r^2 & 0 & 0 & 0 \\ 0 & -1/r^2 & 0 & 0 \\ 0 & 0 & -1/r^2 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

So:

$$R_{\mu\nu} = -\frac{3}{r^2} \eta_{\mu\nu} \quad (\text{Minkowskian metric}).$$

ds spacetime computations

$$ds^2 = -dt^2 + e^{2Ht}(dx^2 + dy^2 + dz^2)$$

$$g_{\mu\nu} = \begin{pmatrix} -1 & & & \\ & e^{2Ht} & & \\ & & e^{2Ht} & \\ & & & e^{2Ht} \end{pmatrix}$$

$$\Gamma_{\mu\nu}^{\rho} = \frac{1}{2} g^{\rho\sigma} (\partial_{\nu} g_{\sigma\mu} + \partial_{\mu} g_{\nu\sigma} - \partial_{\sigma} g_{\mu\nu})$$

$$\Gamma_{\mu\nu}^0 = H e^{2Ht} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\Gamma_{\mu}^1 = H \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\Gamma_{\mu}^2 = H \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\Gamma_{\mu}^3 = H \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

$$R_{\mu\nu} = R^0_{\mu\nu} + R^1_{\mu\nu} + R^2_{\mu\nu} + R^3_{\mu\nu}$$

$$R^0_{\mu\nu} = H^2 e^{2Ht} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$R^1_{\mu\nu} = H^2 \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & e^{2Ht} & 0 \\ 0 & 0 & 0 & e^{2Ht} \end{pmatrix}$$

$$R^2_{\mu\nu} = H^2 \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & e^{2Ht} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & e^{2Ht} \end{pmatrix}$$

$$R^3_{\mu\nu} = H^2 \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & e^{2Ht} & 0 & 0 \\ 0 & 0 & e^{2Ht} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\Rightarrow R_{\mu\nu} = 3H^2 \begin{pmatrix} -1 & & & \\ & e^{2Ht} & & \\ & & e^{2Ht} & \\ & & & e^{2Ht} \end{pmatrix}$$

$$\Rightarrow R_{\mu\nu} = 3H^2 g_{\mu\nu}.$$